FS-CrossMamba: Dual-Branch Frequency-Guided Spatial and Spectral-Attention-Driven Spectral Cross-Mamba Network for Hyperspectral Image Classification

Tianxiang Zhang, Gongcai Zhang, Wenxin Jia, Yuxin Zeng, Mengting Hou, and Jiangyun Li

***Abstract*—The hyperspectral image classification (HSIC) is intrinsically constrained by insignificant interclass discrepancies and high-spectral redundancy. Although Mamba-based methods have appeared recently, current techniques still struggle to properly address the minor differences between classes and substantial spectral redundancy present in HSIC. To overcome these issues, we introduce the dual-branch frequency-guided spatial and spectral-attention-driven spectral Cross-Mamba network (FS-CrossMamba). Unlike conventional methods that estimate state transition parameters solely from the input sequence, our method explicitly injects auxiliary priors into the SSM state transition parameters through two specialized branches: the frequency-domain-guided spatial Cross-Mamba (FSCM) branch, exploiting amplitude and phase priors to guide spatial selective scanning and preserve structural coherence, and the spectral-attention-driven spectral Cross-Mamba (SASCM) branch, utilizing global spectral attention to reweight spectral responses and strengthen long-range dependency modeling. Furthermore, we design a triple-branch convolutional adaptive spectral–spatial coupling module (TCASSCM) to model rich multiscale spectral–spatial relationships as the initial embedding. Comprehensive experiments on five benchmark datasets demonstrate that FS-CrossMamba reliably achieves state-of-the-art results in comparison with top-performing convolutional neural network (CNN)-, Transformer-, and Mamba-based approaches.**

***Index Terms*—Frequency-domain modeling, hyperspectral image classification (HSIC), Mamba, spectral–spatial feature fusion, state-space model (SSM).**

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I. Introduction

HYPERSPECTRAL imaging (HSI) captures continuous and narrow spectral bands across a wide-wavelength range, providing rich reflectance signatures that enable fine-grained material identification beyond the capabilities of conventional RGB or multispectral sensors [1], [2]. Benefiting from its high-spectral resolution and the availability of spatial context, HSI has become a cornerstone of remote sensing applications, including precision agriculture [3], environmental and ecological monitoring [4], urban land-cover mapping [5], and mineral exploration [6]. As a fundamental task in these domains, hyperspectral image classification (HSIC) aims to assign precise semantic labels to pixels by jointly exploiting spectral signatures and spatial contexts. Nonetheless, HSIC consistently faces two inherent data-related challenges: limited separability between classes and pronounced spectral redundancy. To address these challenges, various methods have been proposed, ranging from conventional machine learning techniques to more recent deep learning-based approaches.

Traditional HSIC methods have explored a wide range of techniques based on handcrafted spectral or spectral–spatial features combined with classical classifiers, such as support vector machines (SVMs) [7], [8], [9], k-nearest neighbors (kNNs) [10], [11], [12], and random forests [13], [14]. Dimensionality reduction and spatial enhancement techniques, including principal component analysis (PCA) [15], linear discriminant analysis (LDA) [9], and extended morphological profiles (EMP) [16], have been widely employed to mitigate the spectral redundancy and incorporate spatial context. While these approaches provide effective baselines and demonstrate robustness under limited training samples, their reliance on manually designed feature representations constrains their ability to model complex nonlinear spectral–spatial interactions in challenging scenarios. With the rise of deep learning, HSIC methods have evolved from traditional handcrafted features to data-driven learning. These methods have progressed in three main directions: convolutional neural network (CNN)-based models, Transformer-based models, and Mamba-based models.

Early deep learning approaches primarily relied on CNNs [17], [18]. Notable examples include SSRN [19] and HybridSN [20], which employ 3-D convolutions to capture local spatial–spectral features. Yet, these CNN-based methods

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are not well equipped to address the two core challenges. Spatially, their inherently limited receptive fields fail to capture the global contextual information necessary to discriminate materials with low-interclass separability. Spectrally, because all channels are processed in a uniform manner without an explicit selection mechanism, they are ineffective at suppressing spectral redundancy [21]. To alleviate these locality constraints, Transformers were subsequently introduced to model long-range dependencies; early Transformer-based architectures, such as SpectralFormer [22] and SSFTT [23], have demonstrated considerable promise. Nevertheless, even these Transformer models remain susceptible to the aforementioned issues. On the spatial side, in order to control the quadratic complexity of self-attention, these methods generally rely on aggressive patch partitioning or spatial downsampling. Yet, these strategies inevitably discard fine-grained structural information, leading to blurred object boundaries and thus limiting gains in interclass separability. On the spectral side, their global self-attention modules compute dense pairwise correlations across all bands without explicit filtering, making them vulnerable to spectral redundancy, where noisy or irrelevant bands can be mistakenly treated as informative context [24].

Recently, the Mamba architecture [25] has emerged as a compelling alternative, leveraging state-space models (SSMs) to offer linear computational complexity significantly superior to Transformers. This efficiency has catalyzed a surge of HSI-specific adaptations. On the spatial optimization front, representative models, such as SpectralMamba [26], S2-Mamba [27], and 3DSS-Mamba [28], have introduced various multidirectional scanning strategies to approximate spatial dependencies. Simultaneously, on the frequency enhancement front, approaches like FAHM [29] and WD-SSMamba [30] have integrated Fourier or Wavelet transforms to capture global context. In addition, several recent Mamba-based HSIC studies have explored different strategies to alleviate spectral redundancy and improve class discrimination. For instance, AMamNet [31], HG-Mamba [32], and ASSF-Mamba [33] introduce spectral attention or hierarchical modeling mechanisms to suppress the redundant spectral information and enhance informative band representations. Meanwhile, approaches, such as DADFMamba [34] and BiMambaHSI [35], focus on improving interclass separability by designing more discriminative feature fusion strategies or bidirectional spectral modeling schemes. These studies demonstrate that Mamba-based architectures have shown considerable potential for hyperspectral feature representation and have made meaningful progress in alleviating spectral redundancy or improving class discrimination.

Nevertheless, these advances mainly improve specific aspects of hyperspectral modeling, rather than jointly addressing the two challenges within a unified framework. More specifically, existing methods still exhibit two limitations. From the spatial perspective, although multidirectional scanning and feature fusion can strengthen contextual modeling, they still largely rely on flattening 2-D spatial features into 1-D scanning sequences. This operation inevitably disrupts local neighborhood consistency and weakens the representation of fine geometries and object boundaries, thereby limiting interclass separability. From the spectral perspective, although spectral attention or hierarchical designs can refine spectral features, they usually do not explicitly guide the generation of state transition parameters in the selective SSM. Consequently, an explicit global spectral weighting strategy is still lacking in the generation of state transition parameters, and redundant or uninformative bands may continue to interfere with spectral representation learning. Therefore, existing Mamba-based HSIC methods still leave room for improvement in jointly handling weak interclass separability and high spectral redundancy.

To address these limitations, we propose the dual-branch frequency-guided spatial and spectral-attention-driven spectral Cross-Mamba network (FS-CrossMamba). Distinct from standard Mamba architectures, where state transition parameters are derived solely from the input sequence, FS-CrossMamba uses frequency-enhanced spatial features and attention-weighted spectral representations to dynamically generate the state transition parameters. To implement this design effectively, FS-CrossMamba further incorporates a strong shallow embedding module and two complementary Cross-Mamba branches. Specifically, a triple-branch convolutional adaptive spectral–spatial coupling module (TCASSCM) is first introduced to extract robust multiscale embeddings as the foundational representation. Building on this, the frequency-domain-guided spatial Cross-Mamba (FSCM) branch leverages amplitude and phase priors to generate spatial state transition parameters, effectively preserving structural continuity against 1-D flattening. Simultaneously, the spectral-attention-driven spectral Cross-Mamba (SASCM) branch employs global attention to modulate spectral state transition parameters, thereby filtering spectral redundancy and ensuring discriminative feature learning. By integrating these three core modules within a unified framework, FS-CrossMamba simultaneously addresses the limitations of existing Mamba-based HSIC methods and effectively mitigates the inherent challenges of weak interclass separability and high spectral redundancy. The main contributions of this work are summarized as follows.

1) We propose FS-CrossMamba, a unified framework built on the Mamba model. In contrast to earlier Mamba-based approaches that derive state transition parameters only from the input sequence, FS-CrossMamba employs a dual-branch architecture that injects frequency-enhanced spatial features and attention-refined spectral representations into the SSM dynamic modeling pipeline. This collaborative scheme addresses both subtle interclass differences and spectral redundancy, leading to more robust feature learning and markedly improved the classification performance.
2) We design the TCASSCM module using parallel 1-D, 2-D, and 3-D convolutions to capture diverse yet complementary spectral–spatial information. With an adaptive weighting mechanism, it dynamically fuses multiscale features to enhance the interclass separability,

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forming a strong and discriminative foundation for later cross-domain modeling.

3) We introduce the FSCM branch, applying amplitude–phase decomposition to separate coarse low-frequency structures from fine high-frequency details. These frequency cues are then used to directly steer a spatial Cross-Mamba equipped with a four-direction selective scanning strategy. By injecting complementary frequency information into spatial dynamic modeling, this design significantly enhances semantic representations and improves interclass separability.
4) We design the SASCM branch, employing global spectral attention to highlight informative bands while suppressing the influence of irrelevant wavelengths. A bidirectional spectral Cross-Mamba is then used to capture long-range cross-band dependencies, strengthening discriminative spectral cues and reducing redundancy, thereby improving spectral representation quality and enhancing classification performance.

II. Related Works

The intrinsic challenges of HSIC—marked by subtle interclass variations and substantial spectral redundancy—have shaped the development of feature extraction techniques. Prior studies have primarily aimed to improve the discriminative capability of spectral–spatial representations, while maintaining the computational efficiency. The research trajectory has moved from local feature extraction to global context modeling, and, more recently, toward efficient long-sequence modeling based on SSMs [36], [37], [38].

A. CNN-Based Methods and Transformer-Based Methods

Deep learning has emerged as the leading framework for HSIC, with its development largely driven by CNN- and Transformer-based models. Early studies predominantly used CNNs because of their strong ability to capture local features [21], [39], [40], [41]. Evolving from initial 1-D architectures centered on spectral signatures [42], to 2-D models emphasizing spatial information [43], and ultimately to 3-D architectures [44], CNN-based methods have steadily advanced in modeling joint spectral–spatial representations [45], [46], [47], [48]. Seminal works like HybridSN [20] combined 3-D and 2-D convolutions to strike a balance between capturing fine-grained spectral correlations and learning high-level spatial semantics. Similarly, SSRN [19] introduced residual connections to enable the training of deeper networks. By mitigating the gradient vanishing issue, SSRN supports the learning of highly discriminative hierarchical features, which are essential for separating classes that exhibit only subtle spectral differences between them.

Despite their success, CNNs are intrinsically constrained by the fixed geometry of local receptive fields. This limitation hinders their ability to capture long-range spatial dependencies and global contextual cues effectively [49]. Consequently, their performance often degrades in complex scenes exhibiting significant intraclass variability, and their locality-oriented inductive bias limits their efficacy in disentangling global spectral redundancy. To counter this, Attention-based networks were developed to dynamically weigh the importance of specific spectral channels and spatial regions, effectively suppressing noise and enhancing the representation of the most informative bands [50].

To address the locality limitations inherent in CNNs, Transformers employ a global self-attention mechanism that enables the capture of long-range dependencies over the entire sequence. Numerous Transformer-based models have been specifically designed for HSIC [51], [52], [53], [54], [55], [56], [57]. For instance, SpectralFormer [22], SSFTT [23], and CAT [58] employ strategies such as group-wise spectral embeddings and token reconstruction to better handle the high dimensionality of HSI data. In addition, hybrid frameworks like the cross-attention fusion network [59] combine CNN branches for capturing local textures with Transformer branches for modeling global dependencies, using cross attention to mitigate the disparity between the two modalities.

However, the quadratic computational cost of self-attention with respect to sequence length continues to be a major bottleneck. This limitation makes conventional Transformers less scalable for large hyperspectral scenes or data with very dense spectral sampling, often forcing the use of token pruning or window-based partitioning, which can disrupt global context and compromise the fine-grained details required to distinguish spectrally similar classes.

B. Mamba-Based Methods

Recently, the Mamba architecture has been proposed as a strong alternative for HSIC, utilizing SSMs to model long sequences in linear time [25]. This design is well-suited to the continuous, ordered characteristics of hyperspectral bands, providing an efficient way to model global dependencies without incurring the quadratic computational overhead of Transformers.

Early approaches mostly interpreted spectral vectors as sequential inputs or incorporated Mamba blocks into lightweight backbones [29], [60], [61], [62], [63], [64]. For example, He et al. introduced IGroupSS-Mamba [65] to efficiently model global dependencies in remote sensing images. Similarly, the spectral–spatial Mamba (SS-Mamba) framework [66] constructs distinct spatial and spectral tokens and processes them through Mamba blocks to effectively learn deep feature representations. In addition, DBMGNet [37] adopts a dual-branch design that combines Mamba-based spectral sequence modeling with graph convolutional networks to jointly exploit long-range spectral dependencies and local spatial topology. Recognizing the need for richer representations, recent studies have begun to explore frequency-domain enhancements. Approaches, such as WaveMamba [67] and others [29], [68], utilize Fourier or wavelet transforms to perform spectral filtering, aiming to stabilize long-range interactions and suppress noise. WD-SSMamba [30] combines 1-D and 2-D wavelet decomposition to extract spectral and spatial features in the frequency domain, respectively. It efficiently fuses spectral and spatial features through a specially

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designed dual-branch Mamba block featuring a cross-fusion architecture.

Despite these advances, employing Mamba for HSIC still encounters open issues. Existing recurrence-based designs frequently fail to maintain 2-D spatial continuity because they depend on 1-D scanning schemes. In addition, current frequency-enhanced approaches usually treat frequency transforms as standalone preprocessing or filtering operations, rather than embedding them within a unified framework in which frequency priors (such as amplitude and phase) explicitly steer the spatial scanning procedure. These shortcomings highlight the necessity for a tighter fusion of spatial, spectral, and frequency domains, which in turn motivates the design of the proposed FS-CrossMamba.

III. Methods

This section introduces the proposed FS-CrossMamba framework. We first outline the necessary background and the overall architecture, and then describe in detail the specific designs of its three main modules: TCASSCM, FSCM Branch, and SASCM Branch.

A. Preliminaries

To establish the theoretical basis of the proposed FS-CrossMamba architecture, this section presents the underlying concepts of SSMs and the Mamba architecture. Modern sequence modeling methods based on SSMs can efficiently capture long-range dependencies by mapping a 1-D input signal $x(t) \in \mathbb{R}$ to an output $y(t) \in \mathbb{R}$ via an intermediate latent state vector $\mathbf{h}(t) \in \mathbb{R}^N$. The dynamics of this latent state are governed by the following linear ordinary differential equation (ODE):

$$\begin{aligned} \mathbf{h}'(t) &= \mathbf{A}\mathbf{h}(t) + \mathbf{B}x(t) \\ y(t) &= \mathbf{C}\mathbf{h}(t) + \mathbf{D}x(t) \end{aligned} \tag{1}$$

where $\mathbf{A} \in \mathbb{R}^{N\times N}$ represents the state transition matrix governing the evolution of the latent state, while $\mathbf{B} \in \mathbb{R}^{N\times 1}$ and $\mathbf{C} \in \mathbb{R}^{N\times 1}$ denote the projection matrices for the input and output, respectively, and $\mathbf{D} \in \mathbb{R}^{N\times 1}$ serves as a residual connection that enables a direct mapping from input to output.

To integrate this continuous system into discrete deep learning frameworks, a discretization process is required. A time-scale parameter Δ is introduced to transform the continuous parameters $\mathbf{A}$ and $\mathbf{B}$ into their discrete counterparts, $\overline{\mathbf{A}}$ and $\overline{\mathbf{B}}$, typically employing the zero-order hold (ZOH) technique. This assumes that the input holds constant during the sampling interval Δ

$$\begin{aligned} \overline{\mathbf{A}} &= \exp(\Delta\mathbf{A}) \\ \overline{\mathbf{B}} &= (\Delta\mathbf{A})^{-1}\left(\exp(\Delta\mathbf{A}) - \mathbf{I}\right) \cdot \Delta\mathbf{B}. \end{aligned} \tag{2}$$

From a physical perspective, Δ acts as a sampling interval that governs the granularity of the captured context. A learned, input-dependent Δ allows the model to adaptively focus on fine-grained details (small Δ) or long-range dependencies (large Δ) within the spectral or spatial sequences. Consequently, $\overline{\mathbf{A}}$ determines the retention rate of the previous state information, while $\overline{\mathbf{B}}$ controls the injection of new input information.

Following discretization, the system operates recurrently

$$\begin{aligned} \mathbf{h}_t &= \overline{\mathbf{A}}\mathbf{h}_{t-1} + \overline{\mathbf{B}}x_t \\ y_t &= \mathbf{C}\mathbf{h}_t + \mathbf{D}\mathbf{x}_t \end{aligned} \tag{3}$$

where $\mathbf{h}_{t-1}, \mathbf{h}_t \in \mathbb{R}^N$ denote the hidden states at time steps $t-1$ and t, respectively. In this formulation, $\mathbf{h}_t$ recursively aggregates the historical state $\mathbf{h}_{t-1}$ and current input x_t via discretized transition matrices to project the output y_t. This mechanism, fundamental to SSMs, enables the parameter-efficient modeling of long-range dependencies with linear computational complexity.

This recurrent formulation is realized in the Mamba architecture (Selective SSM) [25] by rendering the parameters $\mathbf{B}$, $\mathbf{C}$, and Δ as input-dependent functions to achieve dynamic context filtering. However, the standard Mamba is insufficient for addressing the substantial intraclass variability, limited interclass separability, and spectral redundancy characteristic of hyperspectral data. To bridge this gap, this article enhances the selective SSM by exploiting spatial–spectral long-range dependencies through four-directional spatial and bidirectional spectral scanning. Furthermore, by incorporating auxiliary features to dynamically derive the state parameters $\mathbf{B}$, $\mathbf{C}$, and Δ, the proposed approach ensures robust feature extraction while maintaining the linear computational complexity.

B. Overall Framework

The overall architecture of the proposed FS-CrossMamba is illustrated in Fig. 1. Let the input HSI be denoted as $\mathbf{I} \in \mathbb{R}^{H\times W\times C}$, where $H \times W$ represents the spatial resolution and C is the number of spectral bands. To capture the local contextual information for pixelwise classification, $\mathbf{I}$ is partitioned into patches $\mathbf{x} \in \mathbb{R}^{S\times S\times C}$ (where $S \times S$ is the patch size). As shown in the pipeline, the input patch $\mathbf{x}$ is first processed by the TCASSCM. This module extracts discriminative multiscale features, denoted by $\mathbf{f} \in \mathbb{R}^{S\times S\times L}$ (typically set to $L = 256$). Subsequently, $\mathbf{f}$ is routed into two parallel branches to model domain-specific dependencies: the frequency-domain-guided spatial Cross-Mamba (FSCM) branch, which focuses on spatial contextualization, and the SASCM branch, which targets spectral correlation. The outputs of these two branches are fused via elementwise addition to yield the deeply integrated representation $\mathbf{Y} \in \mathbb{R}^{S\times S\times L}$. Finally, $\mathbf{Y}$ undergoes global average pooling (GAP) followed by a fully connected layer to generate the final classification predictions.

C. Triple-Branch Convolutional Adaptive Spectral–Spatial Coupling Module

To enhance interclass separability in the shallow layer by capturing diverse spectral–spatial features at multiple scales, we propose the TCASSCM. As depicted in Fig. 1, this module derives complementary priors from the input patch $\mathbf{x} \in \mathbb{R}^{S\times S\times C}$ through three parallel branches. For training stability and to introduce nonlinearity, the operations in each

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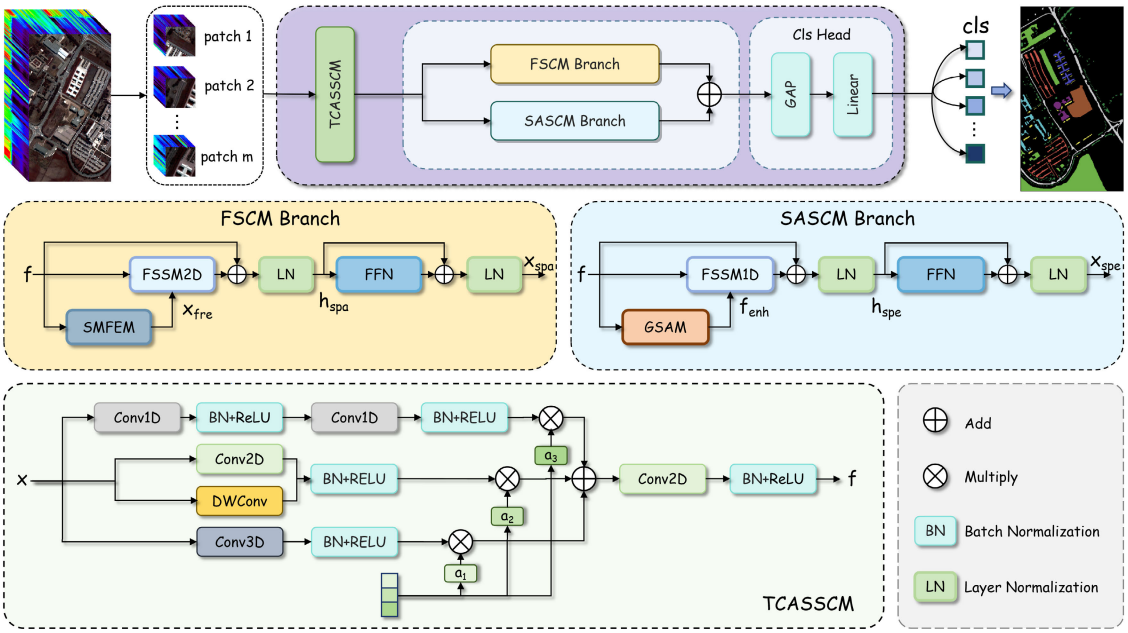


Fig. 1. Overall architecture of the proposed FS-CrossMamba model. The original hyperspectral image is first partitioned into multiple patches and fed into the TCASSCM module to extract shallow spatial–spectral features. These features are then processed independently by the FSCM branch and the SASCM branch, after which the outputs of both branches are integrated. Finally, the fused features pass through a GAP layer and a classifier to produce the final classification results.

branch are followed in sequence by batch normalization and an ReLU activation.

Specifically, the first branch captures the local spatial continuity and sequential dependencies by reshaping the spatial dimensions into a 1-D sequence and applying two successive 1-D convolutions with a kernel size of 3. In parallel, the second branch focuses on extracting local spatial features through a pointwise 1×1 convolution followed by a 3×3 depthwise convolution. The third branch is configured to capture joint volumetric characteristics within local spectral–spatial cubes using a 3×3 3-D convolution.

To merge these multiview representations, the outputs of the individual branches are adaptively combined through a learnable weighting scheme. Specifically, the weight vector $\mathbf{w} \in \mathbb{R}^3$ is used to compute the coefficients $\boldsymbol{\alpha} = (\alpha_1, \alpha_2, \alpha_3)$, which flexibly control the contribution of each branch. The resulting weighted sum is subsequently processed by a 1×1 convolution, producing the final feature map $\mathbf{f} \in \mathbb{R}^{S \times S \times L}$ (with $L = 256$), which is then concurrently processed by the dual-branch interaction framework, consisting of the FSCM and SASCM branches.

D. Frequency-Domain-Guided Spatial Cross-Mamba Branch (FSCM Branch)

To enhance interclass separability by exploiting complementary information in the frequency domain that can reliably steer spatial feature learning, we design the FSCM branch. As illustrated in Fig. 1, this branch strengthens spatial representation learning by combining frequency-domain priors with a selective spatial scanning scheme derived from Fusion SSM2D (FSSM 2D). The FSCM branch operates in a progressive manner, beginning with frequency-guided feature refinement, followed by spatial Cross-Mamba modeling, and concluding with residual feed-forward enhancement.

In particular, the input feature map $\mathbf{f} \in \mathbb{R}^{S \times S \times L}$ is first fed into the spatial Mamba frequency-domain enhancement module (SMFEM). This module explicitly separates the amplitude and phase components in the frequency domain and then reconstructs an improved representation $\mathbf{x}_{\text{fre}}$, which encodes amplified global magnitude information and more precise high-frequency structural details. The resulting frequency-enhanced feature serves as an auxiliary guidance signal that enriches the modeling of spatial dependencies in the subsequent stages.

After the frequency-domain refinement, both $\mathbf{f}$ and $\mathbf{x}_{\text{fre}}$ are fed in parallel into the FSSM 2D module, which conducts a guided spatial cross-fusion. By employing four-directional selective spatial scanning together with state-space parameter modulation, FSSM 2D efficiently models long-range contextual dependencies while exploiting the refined frequency priors. A residual connection is added to stabilize optimization and preserve the original spatial semantics. The resulting intermediate representations are then normalized via layer normalization (LN), yielding

$$\mathbf{h}_{\text{spa}} = \text{LN}\left(\text{FSSM2D}\left(\mathbf{f}, \mathbf{x}_{\text{fre}}\right) + \mathbf{f}\right). \tag{4}$$

To further improve the spatial expressiveness and strengthen the model's ability to generalize, $\mathbf{h}_{\text{spa}}$ is fed into a feed-forward network (FFN), where nonlinear transformations broaden and refine interactions at the channel level. An additional residual connection is incorporated to mitigate over-smoothing and maintain stable gradient flow. The final output of the spatial branch is then obtained as follows:

$$\mathbf{x}_{\text{spa}} = \text{LN}\left(\text{FFN}\left(\mathbf{h}_{\text{spa}}\right) + \mathbf{h}_{\text{spa}}\right). \tag{5}$$

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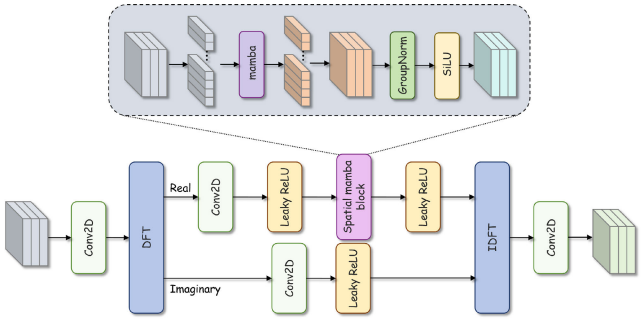


Fig. 2. Proposed SMFEM. Feature $\mathbf{f}$ undergoes 2-D convolution and DFT to obtain amplitude $\mathbf{A}$ and phase $\mathbf{\Phi}$. $\mathbf{A}$ is enhanced via convolution and a spatial Mamba block, while $\mathbf{\Phi}$ is refined by convolution. The fused frequency feature $\widehat{\mathcal{F}}$ is returned to the spatial domain by IDFT and convolved to yield $\mathbf{x}_{\text{fre}}$.

Overall, the FSCM branch effectively integrates frequency-domain guidance, selective spatial scanning, and hierarchical residual refinement. With this design, the network can consistently extract discriminative spatial patterns while maintaining linear computational complexity, making it especially well-suited for hyperspectral spatial modeling.

1) Spatial Mamba Frequency-Domain Enhancement Module: Within the FSCM branch, we integrate the spatial Mamba frequency-domain enhancement module (SMFEM), which provides an explicit and informative prior in the frequency domain to better model spatial dependencies. As shown in Fig. 2, this module decomposes the input feature $\mathbf{f} \in \mathbb{R}^{S\times S\times L}$ into amplitude and phase components in the frequency domain, enhances them separately, and then reconstructs a more expressive spatial representation. First, the input feature map is preprocessed using a 1×1 2-D convolution to adjust its spatial activation distribution. To ensure numerical stability, a small constant ε with a fixed value of 10^{-8} is added before performing the discrete Fourier transform (DFT); the FFT adopts the backward normalization convention, resulting in the complex frequency-domain representation $\mathcal{F}_{\mathbf{f}} \in \mathbb{R}^{S\times(S/2+1)\times L}$. Using an amplitude–phase decomposition strategy, $\mathcal{F}_{\mathbf{f}}$ is then split into its amplitude component $\mathbf{A} = |\mathcal{F}_{\mathbf{f}}| \in \mathbb{R}^{S\times(S/2+1)\times L}$ and phase component $\mathbf{\Phi} = \angle(\mathcal{F}_{\mathbf{f}}) \in \mathbb{R}^{S\times(S/2+1)\times L}$, which are subsequently fed into their respective dedicated branches for processing.

The amplitude branch, which focuses on modeling the global intensity information, first feeds $\mathbf{A}$ into a 2-D convolutional layer, followed by batch normalization and LeakyReLU activation. After obtaining the refined amplitude representation $\mathbf{A}' \in \mathbb{R}^{S\times(S/2+1)\times L}$, the Spatial Mamba module is used to encode long-range magnitude relationships. Meanwhile, the phase branch sequentially applies a 2-D convolution, batch normalization, and LeakyReLU to $\mathbf{\Phi}$, preserving essential structural edges and high-frequency components while suppressing noise.

After these enhancement processes, the updated amplitude and phase components are merged again in the complex frequency domain via Euler's formula to obtain the enhanced spectrum $\widehat{\mathcal{F}} \in \mathbb{R}^{S\times(S/2+1)\times L}$. The signal is then transformed back into the spatial domain using the inverse discrete Fourier transform (IDFT), and its magnitude is taken to guarantee nonnegativity. Finally, a 1×1 2-D convolution is employed to align the channel dimensions, yielding the frequency-enhanced feature $\mathbf{x}_{\text{fre}} \in \mathbb{R}^{S\times S\times L}$, which subsequently serves as a guiding prior for the subsequent FSSM 2D spatial Cross-Mamba modeling.

2) FSSM 2D for Spatial Cross Mamba: In the FSCM branch, spatial dependency modeling is carried out by the FSSM 2D module, which combines the original spatial–spectral feature $\mathbf{f}$ with the frequency-enhanced representation $\mathbf{x}_{\text{fre}}$. Here, the original spatial–spectral feature $\mathbf{f}$ is used as the input sequence $\mathbf{F}_{x2d}$ for spatial selective scanning, while the frequency-enhanced representation $\mathbf{x}_{\text{fre}}$ is taken as $\mathbf{F}_{q2d}$ to guide the generation of the state transition parameters. In this way, the spatial scanning process is directly modulated by the frequency-domain prior. As illustrated in Fig. 3, FSSM 2D first applies rotary positional encoding (RoPE) to the two inputs, $\mathbf{F}_{x2d}$ and $\mathbf{F}_{q2d}$, producing position-aware feature representations. These encoded features are then flattened into sequential forms through four selective spatial scanning schemes: rowwise scanning, columnwise scanning, and their corresponding reversed sequences. This multidirectional scanning enables the model to capture structural patterns and long-range contextual information across the spatial dimensions.

The SSM state-space parameters—namely, the matrices $\mathbf{B}$, $\mathbf{C}$, and the step-size term Δ—are dynamically generated from $\mathbf{F}_{\text{q2d}}$ through trainable linear layers, enabling the module to flexibly adjust its recurrent state transitions based on spatial content. In the SSM computation, $\mathbf{F}_{\text{x2d}}$ is used as the input sequence x_t, ensuring that the enriched frequency information directly influences the selective spatial filtering operation.

The outputs from the four scanning directions are first fused and reshaped into a 2-D structure, producing the final spatially aggregated feature map $\mathbf{F}_{\text{out2d}}$. Algorithm 1 presents the computational procedure of the FSSM 2D. For the sake of clarity, implementation details such as tensor reshaping and permutation are omitted.

Subsequently, the output $\mathbf{F}_{\text{out2d}}$ of FSSM 2D is integrated with the original feature $\mathbf{f}$ in the FSCM through a residual connection, followed by a normalization operation to obtain the intermediate representation $\mathbf{h}_{\text{spa}}$. This representation is then fed into an FFN in the subsequent stage for further feature refinement.

E. Spectral-Attention-Driven Spectral Cross-Mamba Branch (SASCM Branch)

To reduce the spectral redundancy and strengthen discriminative spectral features through a global attention-based modeling strategy, we propose the SASCM branch. As illustrated in Fig. 1, this branch improves spectral discriminability by leveraging global spectral attention along with guided spectral Cross-Mamba modeling. The SASCM branch functions in a sequential manner: it first enhances spectral representations through a global spectral attention module (GSAM), then executes focused spectral scanning via the fusion SSM1D (FSSM 1D) module, and finally employs a residual feed-forward stage for additional refinement.

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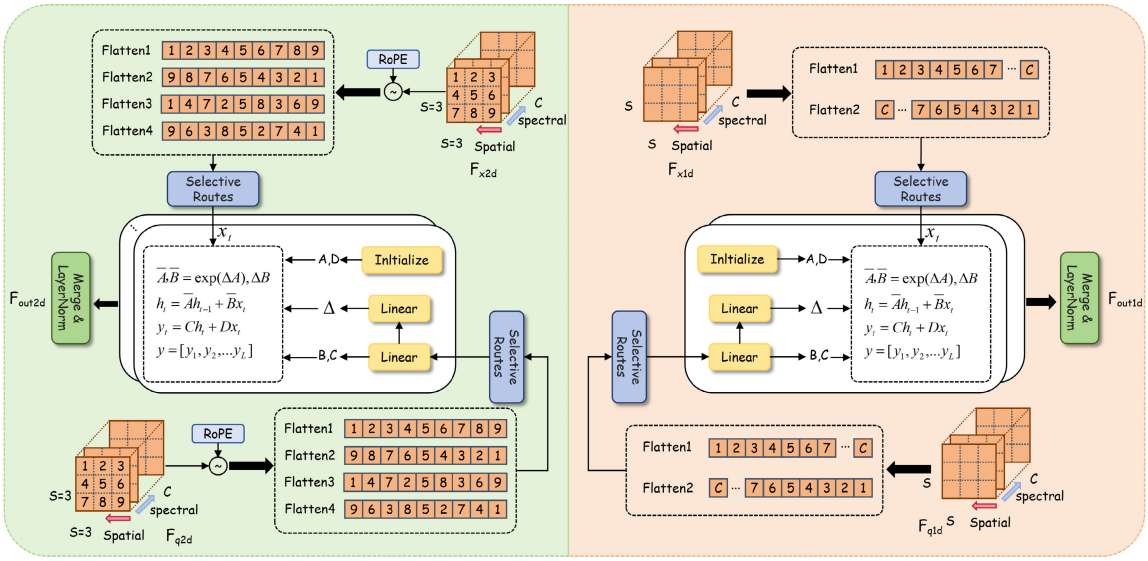


Fig. 3. Proposed fusion modules. The left half depicts the spatial selective scanning and computational process of FSSM 2D, where $\mathbf{F}_{x2d}$ and $\mathbf{F}_{q2d}$ denote the input features and $\mathbf{F}_{\text{out2d}}$ denotes the output feature. The right half presents the channel selective scanning and computation process of FSSM 1D, where $\mathbf{F}_{x1d}$ and $\mathbf{F}_{q1d}$ denote the input features and $\mathbf{F}_{\text{out1d}}$ denotes the output feature.

Algorithm 1 Fusion SSM2D

Require: $F_{x2d}, F_{q2d} \in \mathbb{R}^{S\times S\times L}$
Ensure: $F_{out2d} \in \mathbb{R}^{S\times S\times L}$
where $P = S^2$, $C = L$
Initialize learnable parameters $A \in \mathbb{R}^{C\times N}$, $D \in \mathbb{R}^{C}$
Define spatial serialization operators

$$\mathcal{R} = \{\mathcal{R}_{\text{row}}, \mathcal{R}_{\text{col}}, \mathcal{R}^{\text{rev}}_{\text{row}}, \mathcal{R}^{\text{rev}}_{\text{col}}\}$$

$F_{out2d} \leftarrow 0$
$F_x^{\text{flat}} : (P, C) \leftarrow \text{RoPE}(\text{Flatten}(F_{x2d}))$
$F_q^{\text{flat}} : (P, C) \leftarrow \text{RoPE}(\text{Flatten}(F_{q2d}))$
for all $\mathcal{R}_k \in \mathcal{R}$ **do**
  $F_x^k : (P, C) \leftarrow \mathcal{R}_k(F_x^{\text{flat}})$
  $F_q^k : (P, C) \leftarrow \mathcal{R}_k(F_q^{\text{flat}})$
  $B_k : (P, N) \leftarrow \text{Linear}_B(F_q^k)$
  $C_k : (P, N) \leftarrow \text{Linear}_C(F_q^k)$
  $\Delta_k : (P, C) \leftarrow \text{Softplus}(\text{Linear}_\Delta(F_q^k) + \text{Parameter})$
  $\overline{\mathbf{A}}_k : (P, C, N) \leftarrow \exp(\Delta_k \otimes \mathbf{A})$
  $\overline{\mathbf{B}}_k : (P, C, N) \leftarrow \Delta_k \otimes \mathbf{B}_k$
  $F_k : (P, C) \leftarrow \text{SSM}(\overline{\mathbf{A}}_k, \overline{\mathbf{B}}_k, \mathbf{C}_k, \mathbf{D})(F_x^k)$
  $F_{out2d} : (P, C) \leftarrow F_{out2d} + \mathcal{R}_k^{-1}(F_k)$
end for
$F_{out2d} : (S, S, L) \leftarrow \text{LayerNorm}(F_{out2d})$
return F_{out2d}

Specifically, the input feature map $\mathbf{f} \in \mathbb{R}^{S\times S\times L}$ is first passed through GSAM to produce an attention-enriched representation $\mathbf{f}_{\text{enh}} \in \mathbb{R}^{S\times S\times L}$. GSAM begins by reducing spatial redundancy via adaptive average pooling, followed by Group Normalization to stabilize the resulting feature statistics. The normalized features are then fed into three parallel depthwise separable convolution layers to obtain the query (**Q**), key (**K**), and value (**V**) tensors. These tensors implement a global spectral attention mechanism that aggregates contextual information across all spectral channels. The computed attention map is subsequently compressed with average pooling and passed through a Sigmoid function to form a spectral weighting vector **m**, which is applied to the original features through elementwise multiplication. This process amplifies informative spectral channels while suppressing those that are redundant or contaminated by noise.

After refining the spectral attention, both $\mathbf{f}$ and $\mathbf{f}_{\text{enh}}$ are passed into the FSSM 1D module to carry out guided spectral cross fusion. FSSM 1D begins by flattening the inputs along the spectral axis and applying bidirectional spectral scanning, enabling the model to learn dependencies across consecutive spectral bands. The state-space parameters **B**, **C**, and Δ are adaptively produced from the query features via linear projections, allowing flexible modeling of spectral transitions. The output from the spectral Cross-Mamba operation is subsequently merged with the original features through a residual connection and then normalized with LN, yielding the intermediate representation

$$\mathbf{h}_{\text{spe}} = \text{LN}\left(\text{FSSM1D}\left(\mathbf{f}, \mathbf{f}_{\text{enh}}\right) + \mathbf{f}\right). \tag{6}$$

To enrich the spectral representation and stabilize optimization, $\mathbf{h}_{\text{spe}}$ is passed through an FFN, enabling nonlinear interactions among channels. A second residual connection followed by LN is then applied, yielding the final output of the spectral branch

$$\mathbf{x}_{\text{spe}} = \text{LN}\left(\text{FFN}\left(\mathbf{h}_{\text{spe}}\right) + \mathbf{h}_{\text{spe}}\right). \tag{7}$$

Overall, the SASCM branch combines global spectral attention, bidirectional spectral scanning, and hierarchical residual refinement in a cohesive manner. This architecture enhances the model's ability to capture subtle yet discriminative spectral details while maintaining the computational efficiency, making it particularly suitable for hyperspectral spectral modeling.

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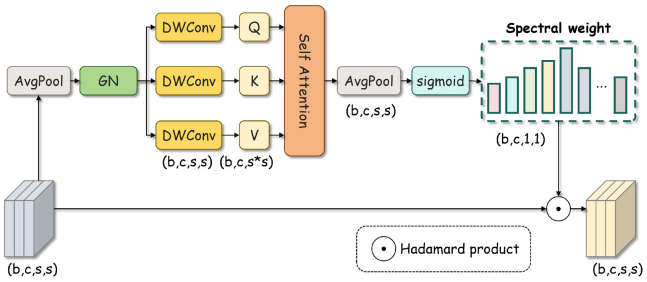


Fig. 4. GSAM. It applies multihead spectral attention to the input feature to extract global spectral dependencies, generating adaptive spectral weights that are fused with the original feature to produce the attention-refined representation $\mathbf{f}_{\text{enh}}$.

1) Global Spectral Attention Module: GSAM is designed to model long-range spectral dependencies and reweight spectral responses, thereby highlighting the most discriminative bands. Acting as the initial stage of the SASCM branch, GSAM adaptively refines the spectral representation of the input feature $\mathbf{f} \in \mathbb{R}^{S\times S\times L}$ prior to spectral Cross-Mamba modeling. As illustrated in Fig. 4, this module applies global spectral attention to capture informative interband relationships, facilitating more effective spectral sequence modeling.

First, the input feature is processed with adaptive average pooling, which compresses spatial redundancy while preserving the inherent spectral properties of each band. The pooled output is subsequently normalized via group normalization to maintain a consistent feature distribution across spectral channels. Afterward, the normalized tensor is projected into the **Q**, **K**, and **V** embeddings using three depthwise separable convolutions. These embeddings are then fed into a GSAM, where interband dependencies are captured by computing the similarity between **Q** and **K** and using the resulting attention weights to aggregate the contextual information from **V**.

The resulting attention output is first compressed using an average pooling operation and then passed through a Sigmoid activation to produce a spectral weighting vector $\mathbf{m} \in \mathbb{R}^{L}$. This vector encodes the overall significance of each spectral band. Finally, **m** is combined with the original input features via elementwise multiplication, yielding an attention-enhanced spectral representation that emphasizes the most informative spectral channels while down-weighting those that are redundant or noisy

$$\mathbf{f}_{\text{enh}} = \mathbf{f} \odot \mathbf{m}. \tag{8}$$

By performing this explicit spectral recalibration, GSAM greatly enhances the model's capacity to discern fine-grained material differences and boosts the robustness of spectral sequence modeling in the subsequent FSSM 1D module.

2) FSSM 1D for Spectral Cross Mamba: In the SASCM branch, spectral dependency modeling is carried out by the FSSM 1D module, which combines the original spectral–spatial feature **f** with the attention-refined representation $\mathbf{f}_{\text{enh}}$. Specifically, the original feature **f** is used as the input sequence $\mathbf{F}_{x1d}$ for spectral scanning, whereas the attention-refined representation $\mathbf{f}_{\text{enh}}$ produced by GSAM is used as $\mathbf{F}_{q1d}$ to guide the generation of the state transition parameters. This design allows the spectral attention prior to be explicitly injected into the spectral state modeling process. As illustrated in Fig. 3, FSSM 1D first flattens the inputs F_{x1d} and F_{q1d} along the spectral dimension, transforming them into sequential spectral representations. This sequential formulation preserves the inherent ordering and continuity of hyperspectral bands within the spectral sequence. Next, two selective spectral scanning schemes—forward and backward traversal—are applied to these sequences, allowing the module to model bidirectional spectral context as well as long-range spectral dependencies.

The state-space parameters of the SSM—namely, **B**, **C**, and the step-size term Δ—are dynamically derived from the query sequence $\mathbf{F}_{\text{q1d}}$ via learnable linear mappings. Through these parameters, FSSM 1D can flexibly adjust its recurrent spectral transition behavior based on the characteristics of the spectral signatures. In the SSM computation, $\mathbf{F}_{\text{x1d}}$ serves as the input sequence x_t, ensuring that the enriched spectral information produced by GSAM is directly incorporated into the selective spectral filtering operation.

The outputs from the forward and backward scanning paths are subsequently fused and reshaped into a 2-D spatial–spectral structure. After LN, the aggregated spectral feature map F_{out1d} is obtained. Algorithm 2 presents the detailed computational procedure of FSSM 1D. For brevity, implementation details, such as tensor reshaping and dimension permutation, are omitted. Subsequently, F_{out1d} is integrated with the original features in the SASCM branch through a residual connection, forming the intermediate representation h_{spe}, which is further refined by an FFN.

IV. Experiment

In this section, we describe the five benchmark datasets and the experimental setup, covering implementation procedures and evaluation metrics, thereby laying the foundation for the subsequent analysis.

A. Experimental Datasets

1) IP Dataset: The Indian Pines (IP) dataset was acquired by the Airborne Visible/Infrared Imaging Spectrometer (AVIRIS) sensor in 1992 over an agricultural site in northwestern Indiana, USA. The image consists of 145 × 145 pixels with a spatial resolution of 20 m. Originally containing 220 spectral bands in the 400–2500-nm wavelength range, bands 104–108, 150–163, and band 220 were removed due to water absorption and low signal-to-noise ratios, leaving 200 valid bands. The dataset comprises 16 land-cover classes, predominantly representing crops and forests. Following the standard protocol defined in SpectralFormer [22], we adopt a fixed training and testing partition. The detailed sample distribution is provided in Table I.

2) PU Dataset: Collected by the Reflective Optics System Imaging Spectrometer (ROSIS) sensor, the Pavia University (PU) dataset captures an urban area over Pavia, Italy. This dataset features a high spatial resolution of 1.3 m and dimensions of 610 × 340 pixels. After discarding 12 noisy bands, 103 spectral bands remain, covering the range from 430 to 860 nm. It includes nine land-cover classes, distinguishing

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Algorithm 2 Fusion SSM1D

Require: $F_{x1d}, F_{q1d} \in \mathbb{R}^{S\times S\times L}$
Ensure: $F_{out1d} \in \mathbb{R}^{S\times S\times L}$
where $P = L$, $C = S^2$
Initialize learnable parameters $\mathbf{A} \in \mathbb{R}^{C\times N}$, $\mathbf{D} \in \mathbb{R}^{C}$
Define channel serialization operators

$$\mathcal{R} = \{\mathcal{R}_{\text{ch}}, \mathcal{R}_{\text{ch}}^{\text{rev}}\}$$

$F_{out1d} \leftarrow 0$
$F_x^{\text{flat}} : (P, C) \leftarrow \text{Flatten}(F_{x1d})$
$F_q^{\text{flat}} : (P, C) \leftarrow \text{Flatten}(F_{q1d})$
for all $\mathcal{R}_k \in \mathcal{R}$ **do**
$F_x^k : (P, C) \leftarrow \mathcal{R}_k(F_x^{\text{flat}})$
$F_q^k : (P, C) \leftarrow \mathcal{R}_k(F_q^{\text{flat}})$
$B_k : (P, N) \leftarrow \text{Linear}_B(F_q^k)$
$C_k : (P, N) \leftarrow \text{Linear}_C(F_q^k)$
$\Delta_k : (P, C) \leftarrow \text{Softplus}(\text{Linear}_\Delta(F_q^k) + \text{Parameter})$
$\overline{\mathbf{A}}_k : (P, C, N) \leftarrow \exp(\Delta_k \otimes \mathbf{A})$
$\overline{\mathbf{B}}_k : (P, C, N) \leftarrow \Delta_k \otimes B_k$
$F_k : (P, C) \leftarrow \text{SSM}(\overline{\mathbf{A}}_k, \overline{\mathbf{B}}_k, \mathbf{C}_k, \mathbf{D})(F_x^k)$
$F_{out1d} : (P, C) \leftarrow F_{out1d} + \mathcal{R}_k^{-1}(F_k)$
end for
$F_{out1d} : (S, S, L) \leftarrow \text{LayerNorm}(F_{out1d})$
return F_{out1d}

TABLE I
TRAINING AND TESTING SAMPLES OF THE IP DATASET

Class No.	Class Name	Training	Testing
1	Corn-notill	50	1384
2	Corn-mintill	50	784
3	Corn	50	184
4	Grass-pasture	50	447
5	Grass-trees	50	697
6	Hay-windrowed	50	439
7	Soybean-notill	50	918
8	Soybean-mintill	50	2418
9	Soybean-clean	50	564
10	Wheat	50	162
11	Woods	50	1244
12	Buildings–Grass–Trees–Drives	50	330
13	Stone–steel towers	50	45
14	Alfalfa	15	39
15	Grass–pasture mowed	15	11
16	Oats	15	5
	Total	**695**	**9671**

between man-made structures (e.g., asphalt and bricks) and natural objects (e.g., meadows and trees). Consistent with the experimental setting in [22], a fixed partition for training and testing is employed, as summarized in Table II.

3) Houston2013 Dataset: The Houston2013 dataset was acquired by the ITRES CASI-1500 sensor over the University of Houston campus, USA. The image features spatial dimensions of 349×1905 pixels with a spatial resolution of 2.5 m. It comprises 144 spectral bands covering the wavelength range from 364 to 1046 nm. This dataset characterizes a complex urban environment with 15 land-cover classes, consisting of

TABLE II
TRAINING AND TESTING SAMPLES OF THE PU DATASET

Class No.	Class Name	Training	Testing
1	Asphalt	548	6304
2	Meadows	540	18146
3	Gravel	392	1815
4	Trees	524	2912
5	Metal Sheets	265	1113
6	Bare Soil	532	4572
7	Bitumen	375	981
8	Bricks	514	3364
9	Shadows	231	795
	Total	**3921**	**40002**

TABLE III
TRAINING AND TESTING SAMPLES OF THE HOUSTON2013 DATASET

Class No.	Class Name	Training	Testing
1	Healthy Grass	198	1053
2	Stressed Grass	190	1064
3	Synthetic Grass	192	505
4	Tree	188	1056
5	Soil	186	1056
6	Water	182	143
7	Residential	196	1072
8	Commercial	191	1053
9	Road	193	1059
10	Highway	191	1036
11	Railway	181	1054
12	Parking Lot 1	192	1041
13	Parking Lot 2	184	285
14	Tennis Court	181	247
15	Running Track	187	473
	Total	**2832**	**12197**

a heterogeneous mixture of man-made structures and natural vegetation. Consistent with the partition utilized in [22], we adopt the fixed training and testing split. The specific sample distribution is detailed in Table III.

4) Trento Dataset: The Trento dataset was captured over a rural area in Trento, Italy, using an AISA Eagle sensor. While LiDAR data are available for this scene, only the hyperspectral data is utilized in this study. The image comprises 600×166 pixels with a spatial resolution of 1 m, containing 63 spectral bands ranging from 402.89 to 989.09 nm. The dataset includes six classes, primarily representing agricultural and rural structures. Following the protocol in [69], distinct training and testing sets are established, as reported in Table IV.

5) WHU-HI-LongKou Dataset: The WHU-HI-LongKou (LK) dataset was acquired using a DJI Matrice 600 Pro UAV platform equipped with a Headwall Nano-Hyperspec sensor in Longkou, Hubei, China. The image features spatial dimensions of 550×400 pixels with a high spatial resolution of approximately 0.46 m. It comprises 270 spectral bands covering the range of 400–1000 nm. Located in an urban–rural fringe area, the scene contains nine land-cover classes. We

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TABLE IV
TRAINING AND TESTING SAMPLES OF THE TRENTO DATASET

Class No.	Class Name	Training	Testing
1	Apples	129	3905
2	Buildings	125	2778
3	Ground	105	374
4	Woods	154	8969
5	Vineyard	184	10317
6	Roads	122	3052
	Total	**819**	**29395**

TABLE V
TRAINING AND TESTING SAMPLES OF THE LONGKOU DATASET

Class No.	Class Name	Training	Testing
1	Corn	54	34457
2	Cotton	53	8321
3	Sesame	54	2977
4	Broad-leaf soybean	53	63159
5	Narrow-leaf soybean	54	4097
6	Rice	50	11804
7	Water	53	67003
8	Roads and houses	53	7071
9	Mixed weed	51	5178
	Total	**475**	**204067**

adopt the standard training and testing split utilized in [29]. The detailed sample distribution is shown in Table V.

B. Experimental Setups

1) Implementation Details: The proposed method was implemented in PyTorch and trained on an NVIDIA GeForce RTX 4090 GPU (24 GB). Experiments were conducted on the five aforementioned HSI datasets. The spatial patch size was fixed at 9×9, and SGD was employed for optimization with a batch size of 32 and weight decay of 0.0011. For different datasets, the learning rate and number of training epochs were set as follows: 0.0009 and 400 for PU and Trento, 0.0007 and 1000 for IP, 0.0004 and 400 for Houston2013, and 0.003 and 400 for LongKou. A step-decay scheduler with a 10-epoch warm-up was adopted to ensure stable convergence and improved generalization. All HSI data were normalized to $[0, 1]$ on a per-band basis, and mirror padding was applied before patch extraction. Standard preprocessed datasets with official training/testing splits were used, with a fixed random seed of 669. For the Mamba-based modules, the state dimension and expansion ratio were set to 16 and 1 for FSSM 2D and FSSM 1D and 16 and 2 for SMFEM. The FFN hidden dimension in both FSCM and SASCM branches was set to 64, with identical input and output dimensions.

2) Evaluation Metrics: To quantitatively evaluate classification performance, we adopt three widely used metrics: overall accuracy (OA), average accuracy (AA), and the Kappa coefficient (κ). OA denotes the fraction of correctly predicted samples out of all test samples, serving as an indicator of overall classification effectiveness. AA is obtained by averaging the classification accuracy over all classes; this measure is especially important for assessing model robustness when dealing with class-imbalanced datasets. The Kappa coefficient quantifies the level of agreement between predicted labels and ground truth while accounting for agreement that might occur by chance. A larger Kappa value reflects higher reliability of the classification outcomes. Together, these three metrics yield a comprehensive assessment that captures OA, per-class performance balance, and statistical agreement.

V. RESULTS AND ANALYSIS

1) *Compared Models:* To assess the effectiveness of the proposed method, nine widely adopted and representative models are chosen as comparison baselines, including SVM [7], 3-D-CNN [44], HybridSN [20], HiT [70], SpectralFormer [22], SSFTT [23], IGroupSS-Mamba [65], FAHM [29], and WTCMC [68]. Among these methods, SVM is a representative of traditional machine learning; 3-D-CNN, HybridSN, and HiT are built on CNN architectures; SpectralFormer and SSFTT employ Transformer architectures; and IGroupSS-Mamba, FAHM, and WTCMC are developed on top of the Mamba architecture. Notably, FAHM and WTCMC additionally embed frequency-domain feature modeling modules into the Mamba framework to further strengthen spectral feature extraction.

A. Comparative Experiments

1) Quantitative Evaluations: Tables VI–X report the quantitative metrics—OA, AA, Kappa, and per-class accuracies—for the proposed FS-CrossMamba method and nine competing techniques on five benchmark HSI datasets. Overall, FS-CrossMamba generally delivers superior performance, attaining the highest OA, AA, and Kappa values in most scenarios, thus highlighting notable gains in both global and class-level recognition.

In scenarios with substantial spectral redundancy, FS-CrossMamba surpasses existing CNN-, Transformer-, and Mamba-based approaches. On the IP dataset, it attains an OA of 97.43%, exceeding the second-best method, FAHM, by 0.72%. Although this OA gain may seem small due to the already saturated performance levels, the consistent improvements in AA (98.62%) and Kappa (0.9705) further demonstrate the robustness of the model, especially in effectively addressing class imbalance. Likewise, on the LongKou dataset, the proposed approach obtains an OA of 95.67%, an AA of 95.47%, and a Kappa of 0.9437, slightly outperforming the runner-up and reinforcing its reliability for large-scale hyperspectral scenes with densely sampled spectral bands.

The performance gains are even more pronounced in high-resolution urban scenes characterized by complex spatial patterns. On the challenging Houston2013 dataset, FS-CrossMamba attains 93.32% OA, 94.32% AA, and 92.74% Kappa, marking a notable improvement of 2.46% in OA over the closest competing method (FAHM: 90.86%). In addition, several particularly challenging categories in Houston2013 are recognized with 100% accuracy, highlighting the

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TABLE VI
COMPARISON OF QUANTITATIVE RESULTS BASED ON THE IP DATASET (IN PERCENTAGE). THE BEST RESULTS ARE HIGHLIGHTED IN BOLD

Class	SVM	3D-CNN	HybridSN	HiT	SpectralFormer	SSFTT	IGroupSS-Mamba	FAHM	WTCMC	Ours
1	67.27	77.38	92.41	80.13	78.83	93.28	93.06	93.21	90.39	**94.65**
2	64.92	70.15	96.17	96.43	85.97	97.58	97.96	99.74	95.92	**99.49**
3	85.87	92.39	98.91	98.37	90.76	**100.00**	**100.00**	**100.00**	**100.00**	**100.00**
4	93.29	94.18	96.87	94.41	95.30	**97.32**	96.87	97.09	95.75	96.64
5	85.94	91.25	99.86	97.27	90.53	**100.00**	99.14	99.86	**100.00**	99.86
6	95.44	98.86	**100.00**	99.09	97.04	**100.00**	99.54	**100.00**	**100.00**	**100.00**
7	75.05	70.37	93.25	82.68	85.29	96.51	91.39	94.77	89.98	**97.93**
8	58.06	65.18	83.62	81.68	73.12	85.03	95.20	95.33	90.82	**96.48**
9	78.72	70.04	93.97	85.46	74.29	92.38	**97.34**	94.33	88.65	94.33
10	98.77	98.15	**100.00**	**100.00**	**100.00**	**100.00**	**100.00**	**100.00**	**100.00**	**100.00**
11	87.54	90.59	98.47	93.65	92.85	97.19	**98.95**	98.79	98.71	98.47
12	65.76	92.12	98.18	**100.00**	89.70	98.79	99.70	**100.00**	**100.00**	**100.00**
13	95.56	97.78	**100.00**	**100.00**	**100.00**	**100.00**	**100.00**	**100.00**	**100.00**	**100.00**
14	82.05	94.87	97.44	97.44	92.31	**100.00**	**100.00**	**100.00**	**100.00**	**100.00**
15	90.91	**100.00**	**100.00**	**100.00**	90.91	**100.00**	**100.00**	**100.00**	**100.00**	**100.00**
16	**100.00**	**100.00**	**100.00**	**100.00**	**100.00**	**100.00**	**100.00**	**100.00**	**100.00**	**100.00**
OA(%)	73.68	78.44	93.07	88.44	83.69	93.78	96.30	96.71	94.02	**97.43**
AA(%)	82.82	87.71	96.82	94.16	89.81	97.35	98.07	98.32	96.89	**98.62**
$\kappa \times 100$	70.27	75.51	92.09	86.82	81.45	92.90	95.75	96.23	93.16	**97.05**

TABLE VII
COMPARISON OF QUANTITATIVE RESULTS BASED ON THE PU DATASET (IN PERCENTAGE). THE BEST RESULTS ARE HIGHLIGHTED IN BOLD

Class	SVM	3D-CNN	HybridSN	HiT	SpectralFormer	SSFTT	IGroupSS-Mamba	FAHM	WTCMC	Ours
1	75.02	80.96	90.13	87.93	78.47	92.02	**98.24**	94.34	89.86	96.75
2	68.60	93.09	69.69	73.34	83.72	75.28	72.19	96.25	68.06	**96.95**
3	72.34	69.26	78.29	72.95	62.92	84.52	75.04	88.60	77.25	**89.53**
4	90.93	97.29	77.06	88.80	92.41	88.19	93.72	**97.66**	97.08	97.60
5	99.19	**100.00**	**100.00**	99.73	99.28	99.73	99.37	99.10	**100.00**	99.28
6	91.23	61.35	97.97	**99.91**	63.74	91.19	98.84	98.38	92.10	96.43
7	87.46	78.80	96.64	84.71	72.58	97.76	95.51	98.98	97.15	**99.59**
8	88.56	94.26	97.12	93.31	94.59	86.09	96.43	**98.75**	90.22	98.10
9	**100.00**	90.82	97.11	94.59	96.73	96.73	99.62	97.61	99.50	98.87
OA(%)	77.61	86.67	81.43	82.90	81.63	83.66	84.95	96.33	80.87	**96.84**
AA(%)	85.93	85.09	89.33	88.36	82.72	90.17	92.11	96.63	90.14	**97.01**
$\kappa \times 100$	71.57	81.95	76.38	78.10	75.63	78.92	80.76	95.08	75.68	**95.74**

clear superiority of our frequency-guided spatial modeling in addressing strong intraclass variability, where alternative approaches underperform.

FS-CrossMamba also exhibits strong generalization across different types of scenes, maintaining a clear performance edge on both the urban PU and rural Trento benchmarks. On the PU dataset, the proposed approach attains 96.84% OA, 97.01% AA, and 95.74% Kappa. On the Trento dataset, it achieves 98.06% OA, 97.06% AA, and 97.40% Kappa, consistently surpassing FAHM. This stable advantage in both urban and rural environments indicates that the proposed framework can reliably capture discriminative features independent of the specific scene semantics.

In conclusion, although performance on certain saturated datasets improves only incrementally, the substantial leap achieved on the challenging Houston2013 dataset, together with consistently state-of-the-art outcomes across diverse scenes, demonstrates that FS-CrossMamba exhibits the strong generalization capability. Through the synergistic combination of spectral attention, frequency-domain guidance, and CrossMamba modeling, the proposed approach sets a new standard for HSIC.

2) Qualitative Results: Figs. 5–9 present the classification maps for the IP, PU, Houston2013, Trento, and LongKou datasets, respectively. For comparison, a false-color composite image and its corresponding ground-truth map are shown for each scene. A visual examination indicates that the qualitative outcomes are highly consistent with the quantitative metrics reported in Tables VI–X, thereby confirming the reliability of the proposed method.

Across all benchmark datasets, FS-CrossMamba produces the most visually coherent classification maps, exhibiting improved spatial smoothness and very low noise. In agricultural scenes such as IP, it effectively reduces salt-and-pepper noise, preserving broad homogeneous regions while accurately tracing the boundaries between neighboring crop fields. A similar pattern appears in the Trento dataset, where FS-CrossMamba maintains clear, continuous borders among vineyards, roads, and buildings. In particular, it reconstructs road networks with greater structural accuracy than rival methods, preserving connectivity and avoiding fragmentation.

The model's strong spatial modeling ability generalizes well to complex urban scenes. On the PU dataset, FS-CrossMamba faithfully maintains the geometric outlines of

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TABLE VIII

COMPARISON OF QUANTITATIVE RESULTS BASED ON THE HOUSTON2013 DATASET (IN PERCENTAGE). THE BEST RESULTS ARE HIGHLIGHTED IN BOLD

Class	SVM	3D-CNN	HybridSN	HiT	SpectralFormer	SSFTT	IGroupSS-Mamba	FAHM	WTCMC	Ours
1	83.00	81.77	82.24	82.05	83.00	86.04	**92.31**	83.67	83.10	83.48
2	98.40	93.52	85.34	97.74	96.80	97.74	98.97	98.59	93.52	**100.00**
3	99.21	82.57	97.82	83.96	71.88	**100.00**	99.60	99.60	97.62	99.60
4	98.20	96.88	80.97	96.50	93.94	**98.39**	95.36	98.20	94.98	97.82
5	97.73	98.77	**100.00**	99.34	98.58	**100.00**	**100.00**	**100.00**	**100.00**	**100.00**
6	79.02	81.12	95.80	89.51	91.61	97.20	**100.00**	**100.00**	**100.00**	98.60
7	65.95	87.50	68.56	89.55	91.51	78.26	75.93	**94.87**	82.46	94.59
8	53.47	81.67	79.68	86.70	78.25	88.03	75.97	80.82	**88.41**	87.27
9	65.63	81.40	87.72	77.71	78.47	89.71	**95.75**	84.51	80.08	76.30
10	37.36	56.18	74.71	40.64	66.41	67.37	72.01	87.16	82.82	**97.68**
11	74.57	75.43	91.37	90.61	64.23	96.39	93.36	81.02	**96.68**	96.39
12	51.59	78.00	93.18	78.39	76.56	**93.08**	92.22	91.35	91.35	91.93
13	39.65	80.70	77.54	78.95	65.61	**96.49**	87.37	78.60	91.93	91.58
14	97.57	98.38	**100.00**	99.60	87.85	**100.00**	**100.00**	99.19	**100.00**	99.60
15	96.83	75.69	99.79	81.61	78.01	97.89	**100.00**	**100.00**	99.79	**100.00**
OA(%)	74.53	83.09	85.81	84.19	82.00	90.75	90.36	90.86	90.49	**93.32**
AA(%)	75.88	83.30	87.65	84.86	81.52	92.44	91.92	91.84	92.18	**94.32**
$\kappa \times 100$	72.36	81.63	84.59	82.83	80.45	89.97	89.53	90.08	89.67	**92.74**

TABLE IX

COMPARISON OF QUANTITATIVE RESULTS BASED ON THE TRENTO DATASET (IN PERCENTAGE). THE BEST RESULTS ARE HIGHLIGHTED IN BOLD

Class	SVM	3D-CNN	HybridSN	HiT	SpectralFormer	SSFTT	IGroupSS-Mamba	FAHM	WTCMC	Ours
1	87.07	98.57	98.75	96.08	92.60	97.08	**100.00**	99.31	77.90	99.82
2	94.60	84.34	76.64	82.87	94.35	92.80	90.03	**96.90**	58.78	94.38
3	95.72	71.93	79.14	74.33	60.70	93.85	92.51	98.66	87.70	**99.47**
4	95.90	97.61	93.47	98.68	84.60	99.53	**99.64**	99.29	97.94	98.99
5	70.73	99.53	54.43	94.92	97.17	82.90	88.40	99.99	84.11	**100.00**
6	46.85	70.61	80.60	70.87	55.93	80.18	80.80	82.37	89.02	**89.71**
OA(%)	80.67	94.03	77.36	92.33	87.72	90.65	92.79	97.55	85.67	**98.06**
AA(%)	81.81	87.10	80.51	86.29	80.89	91.06	91.90	96.09	82.58	**97.06**
$\kappa \times 100$	74.64	91.99	71.25	89.76	83.72	87.69	90.45	96.71	80.91	**97.40**

TABLE X

COMPARISON OF QUANTITATIVE RESULTS BASED ON THE LONGKOU DATASET (IN PERCENTAGE). THE BEST RESULTS ARE HIGHLIGHTED IN BOLD

Class	SVM	3D-CNN	HybridSN	HiT	SpectralFormer	SSFTT	IGroupSS-Mamba	FAHM	WTCMC	Ours
1	85.16	93.87	89.08	97.50	83.76	94.70	95.53	**99.52**	97.85	98.85
2	27.76	84.67	88.27	62.71	50.64	96.08	96.71	**99.25**	96.86	98.44
3	53.21	72.19	98.52	80.65	53.81	97.55	97.45	88.68	**99.83**	85.72
4	69.81	76.04	54.92	80.61	83.90	72.09	**89.64**	87.35	75.90	89.18
5	81.57	76.47	90.65	64.07	70.10	96.24	97.19	**99.17**	98.19	97.61
6	84.22	80.16	72.35	**99.25**	81.71	92.40	94.33	**99.25**	94.36	**99.25**
7	**99.99**	98.40	95.89	98.81	99.95	99.30	98.57	99.55	99.78	99.56
8	83.84	92.69	59.27	93.73	80.46	76.00	90.17	91.26	81.88	**95.04**
9	42.93	92.20	75.11	84.20	74.10	83.43	88.84	95.17	88.62	**95.56**
OA(%)	81.23	87.92	78.52	90.00	86.58	88.28	94.39	95.18	90.70	**95.67**
AA(%)	69.83	85.19	80.45	84.61	75.38	89.75	94.27	**95.47**	92.59	**95.47**
$\kappa \times 100$	76.12	84.52	72.58	87.12	82.61	85.02	92.71	93.75	88.08	**94.37**

artificial structures. For the challenging Houston2013 dataset, it effectively separates classes with highly similar spectra. In particular, it reliably captures continuous linear features, such as "highway" areas, and distinguishes fine-scale structures like the "running track," while competing approaches often suffer from fragmented predictions at the pixel level. In addition, on the LongKou dataset, FS-CrossMamba exhibits exceptional robustness to complex spatial layouts, delivering smooth, consistent classifications of narrow, closely spaced crop strips with minimal noise contamination.

To further assess the discriminative capacity of the learned representations, we employ t-distributed stochastic neighbor embedding (t-SNE) to visualize the feature embeddings produced by different approaches, as shown in Fig. 10. As can be seen, the feature distributions of the competing methods generally display loose intraclass clusters and substantial interclass overlap, revealing limited separability. In comparison, FS-CrossMamba establishes a well-organized feature manifold with tightly grouped samples within each class and clearly separated margins between classes. This result verifies that,

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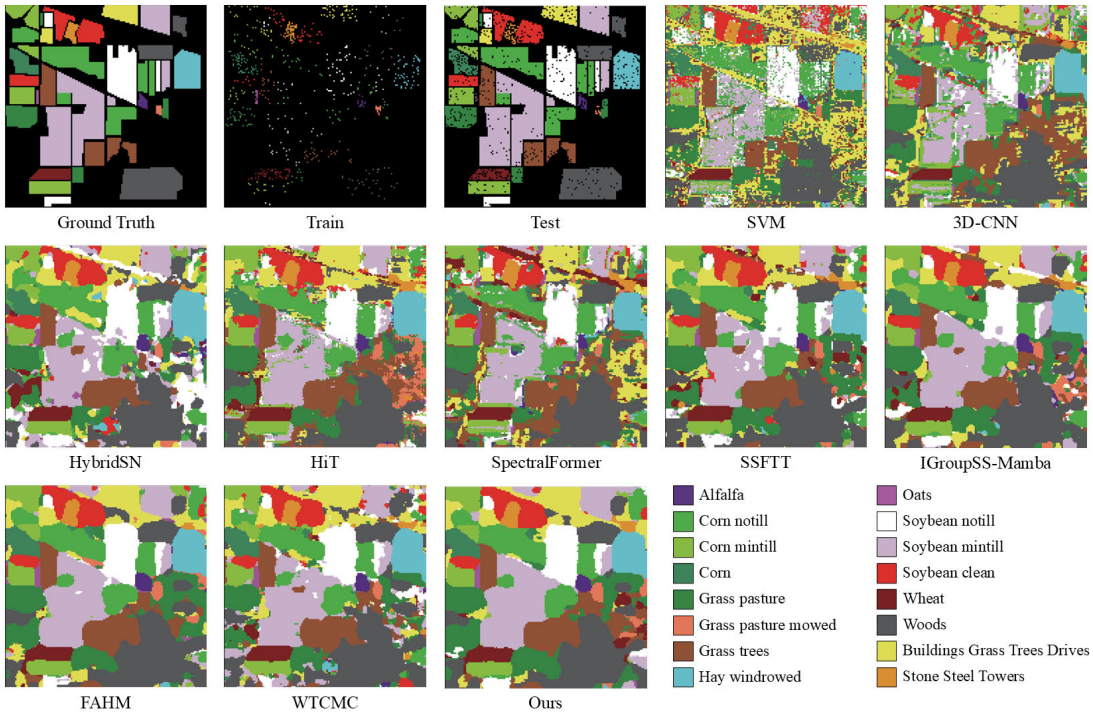


Fig. 5. Classification maps obtained by different models on the IP dataset. Note that the black regions in the ground truth represent unlabeled areas.

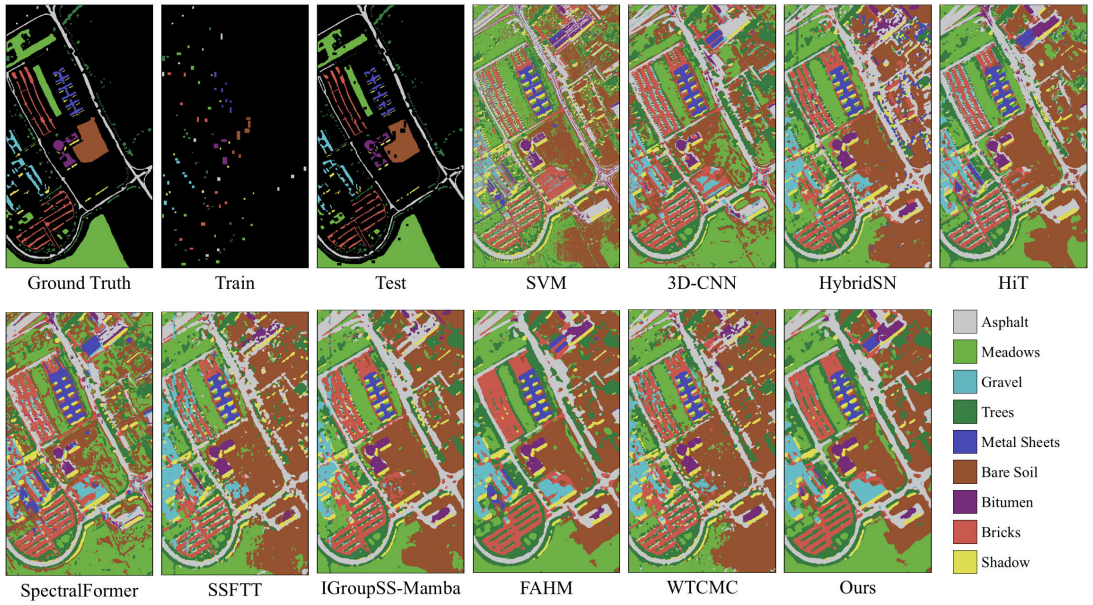


Fig. 6. Classification maps obtained by different models on the PU dataset. Note that the black regions in the ground truth represent unlabeled areas.

by jointly leveraging frequency-domain cues and spectral information, the proposed framework learns high-level features that are simultaneously discriminative and robust.

In conclusion, the superior visual clarity of the classification maps, consistent with the improved feature separability observed via t-SNE, indicates that FS-CrossMamba successfully mitigates classification noise while maintaining the structural coherence. These qualitative results offer strong interpretative support for the state-of-the-art quantitative performance attained by the proposed approach.

B. Parameter Analysis

In this section, we conduct an extensive sensitivity analysis of the main hyperparameters—namely, spatial patch size, training batch size, and feature embedding dimension—to evaluate their influence on classification performance and identify the optimal settings for the proposed FS-CrossMamba.

1) Impact of Patch Size: To determine the spatial receptive field that optimally balances contextual coverage with computational cost and robustness to noise, we evaluated five patch sizes, $\{5 \times 5, 7 \times 7, 9 \times 9, 11 \times 11, 13 \times 13\}$, on the

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TABLE XI
CLASSIFICATION PERFORMANCE UNDER DIFFERENT PATCH SIZES ON FOUR DATASETS (IN PERCENTAGE)

Patch Sizes	Indian Pines			Pavia University			Houston2013			Trento		
	OA (%)	AA (%)	Kappa	OA (%)	AA (%)	Kappa	OA (%)	AA (%)	Kappa	OA (%)	AA (%)	Kappa
5 × 5	92.21	96.02	91.10	92.73	94.21	90.29	90.81	92.20	90.02	96.94	95.25	95.90
7 × 7	97.02	98.49	96.59	94.92	96.32	93.21	92.19	93.51	91.52	97.64	96.50	96.84
9 × 9	**97.43**	**98.62**	**97.05**	**96.84**	**97.01**	**95.74**	93.32	**94.32**	92.74	**98.06**	**97.06**	**97.40**
11 × 11	96.74	98.14	96.27	96.84	95.62	95.75	**93.42**	94.10	**92.86**	96.95	94.53	95.91
13 × 13	95.58	97.07	94.93	96.42	96.27	95.18	91.92	92.67	91.23	96.94	95.26	95.90

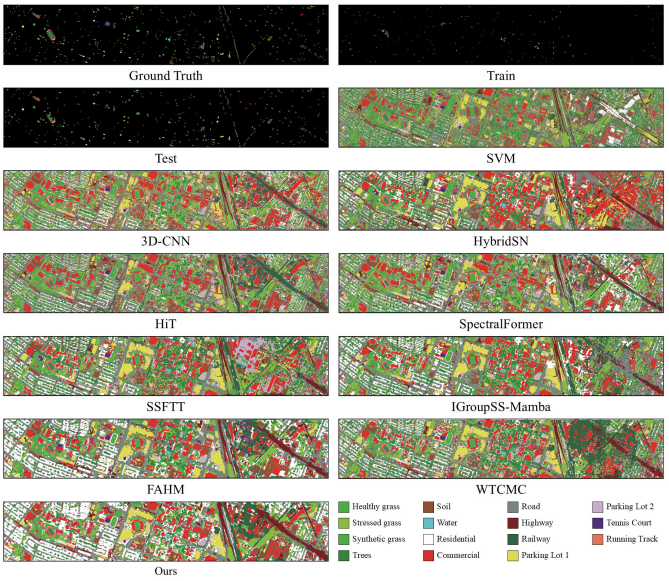


Fig. 7. Classification maps obtained by different models on the Houston dataset. Note that the black regions in the ground truth represent unlabeled areas.

IP, PU, Houston2013, and Trento datasets. The corresponding quantitative results are presented in Table XI. We find that the IP, PU, and Trento datasets achieve their highest OA using a 9 × 9 patch size. Although Houston2013 attains slightly higher accuracy with 11 × 11, the gain over 9 × 9 is minimal and does not warrant the extra computational cost. Consequently, we select a patch size of 9×9 for all subsequent experiments.

2) Impact of Batch Size: The batch size directly influences both the reliability of gradient estimates and the convergence path during optimization. To investigate this impact, we examine four batch sizes, {16, 32, 64, 128}, on the PU and Houston2013 datasets. As shown in Fig. 11(a) and (b), the training behavior varies markedly with batch size. Using a small batch size introduces substantial variance in the gradient estimates, which in turn causes unstable optimization and slower convergence. In contrast, an excessively large batch size diminishes gradient stochasticity and can drive the model toward sharp local minima, thereby harming generalization. Among the evaluated configurations, a batch size of 32 achieves the best OA, AA, and Kappa scores, suggesting it strikes a favorable balance between gradient stability and generalization capability. Therefore, we adopt a batch size of 32 in our experiments.

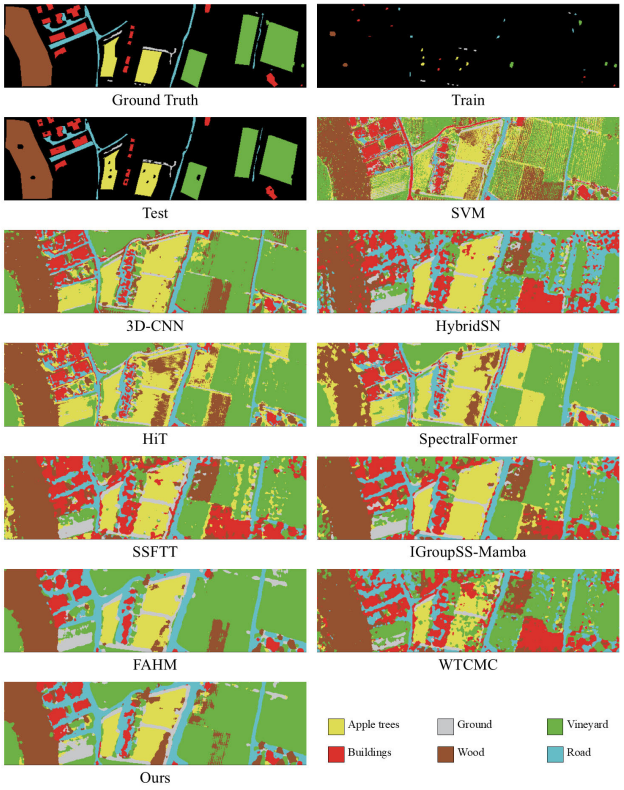


Fig. 8. Classification maps obtained by different models on the Trento dataset. Note that the black regions in the ground truth represent unlabeled areas.

3) Impact of Feature Dimension: The dimensionality of the feature embeddings determines how well the model can capture complex spectral–spatial relationships. We evaluate four channel settings, {128, 196, 256, 320}, on the PU and Houston2013 datasets, with the corresponding results presented in Fig. 11(c) and (d). The performance follows a distinctly nonlinear pattern: when the dimensionality is too low, an information bottleneck arises, which limits the expressiveness of the learned features. Conversely, overly high dimensionality introduces redundant features, heightening the risk of overfitting and increasing computational cost. The experiments show that a dimension of 256 yields the best accuracy, striking a suitable compromise between representational power and model complexity. Therefore, we adopt 256 as the feature dimension.

C. Ablation Study

1) Ablation Study on TCASSCM: To evaluate the impact of TCASSCM, we performed an ablation study in which

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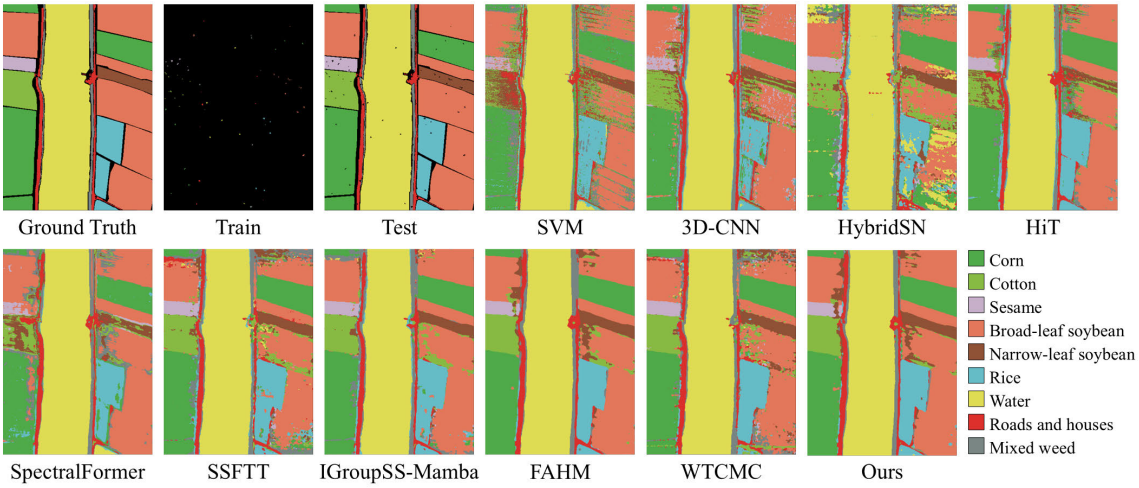


Fig. 9. Classification maps obtained by different models on the LongKou dataset. Note that the black regions in the ground truth represent unlabeled areas.

TABLE XII

ABLATION RESULTS. CHECKMARK (✓) INDICATES THE CORRESPONDING MODULE IS ENABLED. BOLD FONT HIGHLIGHTS THE BEST (FULL MODULE) RESULTS

Modules			Indian Pines			Pavia University			Trento			Houston2013		
TCASSCM	FSCM	SASCM	OA(%)	AA(%)	$\kappa \times 100$	OA(%)	AA(%)	$\kappa \times 100$	OA(%)	AA(%)	$\kappa \times 100$	OA(%)	AA(%)	$\kappa \times 100$
✓			97.07	98.48	96.64	95.59	96.41	94.56	97.82	97.00	97.07	90.87	92.03	90.09
✓		✓	95.50	97.98	94.85	91.98	91.97	89.28	96.27	88.48	94.99	92.81	93.99	92.19
✓	✓		97.35	98.57	96.96	96.39	95.61	95.12	97.94	96.95	97.23	92.51	93.46	91.87
	✓	✓	97.23	98.45	96.82	92.14	90.45	89.17	97.12	95.96	96.14	92.24	92.92	91.57
✓	✓	✓	**97.43**	**98.62**	**97.05**	**96.84**	**97.01**	**95.74**	**98.06**	**97.06**	**97.40**	**93.32**	**94.32**	**92.74**

TCASSCM was used as the only feature extractor, and all subsequent cross-domain Mamba branches were disabled. As reported in Table XII, when relying solely on TCASSCM, the model still attains strong performance on the IP dataset, achieving 97.07% OA, 98.48% AA, and a Kappa of 0.9664. This demonstrates that TCASSCM can effectively derive discriminative shallow features through multiview spectral–spatial coupling. In contrast, when TCASSCM is removed, the performance becomes suboptimal even with the advanced cross-domain branches enabled (see the configuration without TCASSCM), implying that accurate modeling of long-range dependencies strongly depends on high-quality initial feature priors. Therefore, TCASSCM functions as a crucial front-end component that both stabilizes the optimization process and improves feature discriminability by capturing complementary multiscale spectral–spatial information.

2) Ablation Study on FSCM Branch: We additionally examine the contribution of the FSCM branch. A comparison between the baseline model (using only TCASSCM) and the TCASSCM + FSCM configuration shows consistent improvements across all datasets. On the PU dataset, for example, the OA rises from 95.59% to 96.39%, while the Kappa coefficient increases from 0.9456 to 0.9512. Comparable gains appear on the Houston2013 dataset (90.87% → 92.51% OA) and the Trento dataset (97.82% → 97.94% OA). These observations indicate that the FSCM branch substantially strengthens the modeling of long-range spatial dependencies. By combining frequency-domain priors with guided spatial modeling, this branch enhances spatial coherence and increases robustness to structural ambiguity, especially in complex scenes.

3) Ablation Study on SASCM Branch: Finally, we assess the SASCM branch. Notably, activating SASCM in isolation together with TCASSCM (denoted as the TCASSCM-SASCM configuration) does not reliably improve performance and can even cause degradation in certain cases (for instance, PU: 95.59% → 91.98% OA). This indicates that spectral dependency modeling on its own cannot adequately capture complex scene structures without sufficient spatial guidance. In contrast, the complete model (combining TCASSCM, FSCM, and SASCM) delivers the best results across all four datasets, achieving 96.84% OA on PU and 93.32% OA on Houston2013. It highlights that our model is a systematic integration rather than a simple additive collection of modules. These findings demonstrate a pronounced synergistic effect: the SASCM branch enhances the FSCM branch by improving spectral discriminability (through GSAM and FSSM 1D), and its benefits are fully realized only when paired with strong spatial modeling. This supports the necessity of the proposed dual-branch cross-domain architecture.

4) Ablation Study on the SMFEM: To further assess the impact of SMFEM, we conducted an ablation study by removing it from the model. As shown in Table XIII, performance declines consistently across all datasets. For example, on the IP dataset, OA drops to 96.83% (a decrease of 0.60%), and the

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TABLE XIII

ABLATION-STYLE COMPARISON OF GSAM, SMFEM, AND THEIR COMBINATION ON FOUR HYPERSPECTRAL DATASETS. ENABLED MODULES ARE MARKED WITH A CHECK (✓), AND THE BEST RESULTS ARE SHOWN IN BOLD

Modules		Indian Pines			Pavia University			Trento			Houston2013		
SMFEM	GSAM	OA(%)	AA(%)	$\kappa \times 100$	OA(%)	AA(%)	$\kappa \times 100$	OA(%)	AA(%)	$\kappa \times 100$	OA(%)	AA(%)	$\kappa \times 100$
✓		97.27	98.54	96.87	95.88	96.71	94.48	97.76	96.65	97.00	91.46	92.97	90.72
	✓	96.83	98.25	96.36	93.08	94.41	90.78	97.49	96.26	96.64	92.24	93.08	91.57
✓	✓	**97.43**	**98.62**	**97.05**	**96.84**	**97.01**	**95.74**	**98.06**	**97.06**	**97.40**	**93.32**	**94.32**	**92.74**

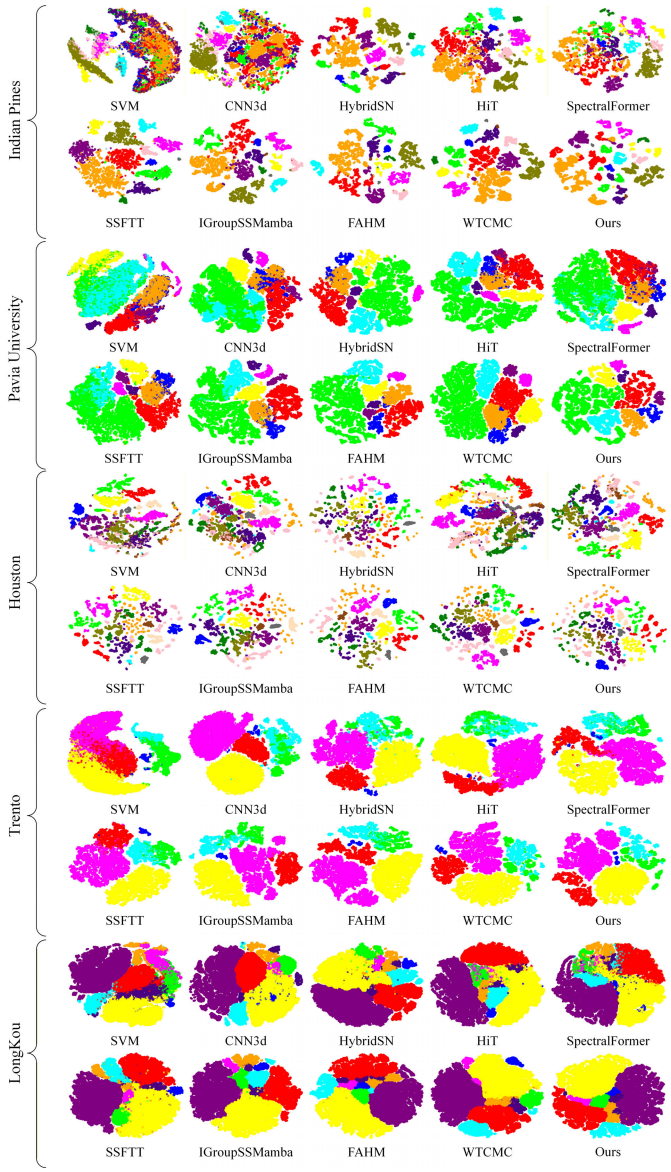


Fig. 10. Feature distribution (t-SNE visualization) obtained by different methods of five datasets.

Kappa coefficient decreases from 0.9705 to 0.9636. Similar trends are observed on other datasets. SMFEM effectively strengthens the structural and textural representations, allowing the model to capture critical spatial patterns and provide informative priors, thereby enhancing the modeling of spatial dependencies and improving classification performance.

5) Ablation Study on the GSAM: We also evaluate the contribution of GSAM through ablation experiments. Excluding GSAM results in decreased performance across all datasets. For instance, on the PU dataset, OA drops to 95.88% (a decrease of 0.96%), and the Kappa coefficient decreases from 0.9574 to 0.9448. Similar trends are observed on the IP and other datasets. GSAM selectively emphasizes spectrally informative bands while suppressing redundant responses, modeling dependencies among spectral channels, and thereby enhancing overall model performance.

D. Analysis of Computational Efficiency

Table XIV presents a comparison of inference time, parameter count (Params), and floating point operations (FLOPs) between the proposed method and several state-of-the-art approaches across five datasets. Here, inference time refers to the total runtime required to classify all pixels in an entire hyperspectral image. The evaluation was conducted under a unified testing setup with a batch size of 100 and FP32 precision on an NVIDIA GeForce RTX 4090 GPU. All preprocessing operations, including normalization and image patch extraction, were completed prior to inference and are therefore excluded from the reported times. Although deep learning-based methods naturally demand more computation than traditional SVM, our approach preserves a favorable trade-off between efficiency and accuracy. Relative to Transformer-based models such as SpectralFormer (e.g., 2.88 s inference on IP) and HiT (3.92 s), our method (1.57 s) runs considerably faster, benefiting from the linear-time complexity of the Mamba architecture. While lightweight CNNs like SSFTT contain fewer parameters, they often fail to adequately capture long-range dependencies, which lead to reduced accuracy. Consequently, although SSFTT achieves a shorter inference time (0.56 s on IP), its lightweight design limits its ability to model complex spectral–spatial interactions. In contrast, the proposed FS-CrossMamba introduces additional cross-dimensional feature modeling and more expressive representation learning modules, which slightly increases the computational cost but significantly improves the classification accuracy. Notably, our method demonstrates a clear efficiency advantage over other Mamba-based models, such as IGroupSS-Mamba (6.08 s on IP). Overall, the proposed method maintains competitive computational efficiency while achieving consistently superior classification performance across the evaluated datasets. Although the proposed FS-CrossMamba slightly increases the parameter count relative to 3-D-CNN, it delivers notable performance

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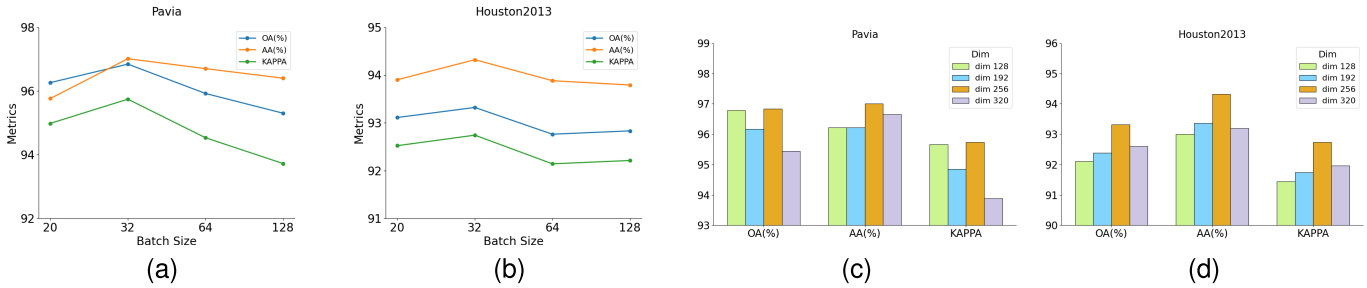


Fig. 11. Impact of batch size and feature dimension on the training dynamics of the model on the PU and Houston2013 datasets. (a) Effect of batch size on the PU dataset. (b) Effect of batch size on the Houston2013 dataset. (c) Effect of feature dimension on the PU dataset. (d) Effect of feature dimension on the Houston2013 dataset.

TABLE XIV

COMPARISON OF INFERENCE TIME, PARAMETER AMOUNT, AND FLOPS ON FIVE DATASETS. THE BEST RESULTS ARE SHOWN IN BOLD

Dataset	Metric	SVM	3D-CNN	HybridSN	HiT	SpectralFormer	SSFTT	IGroupSS-Mamba	FAHM	WTCMC	Ours
Indian	Inference (s)	0.6	1.93	**0.5**	3.92	2.88	0.56	6.08	1.83	4.8	1.57
	Params (M)	–	0.395	0.404	59.286	1.043	0.153	**0.136**	1.077	0.304	1.289
	FLOPs (G)	–	0.069	0.006	0.427	0.044	**0.004**	0.005	0.082	0.008	0.053
Pavia	Inference (s)	16.9	5.9	**4.83**	32.68	16.12	5.45	57.69	17.26	16.3	13.66
	Params (M)	–	0.141	0.403	54.505	0.444	0.153	0.136	0.681	**0.131**	1.025
	FLOPs (G)	–	0.036	0.006	0.329	0.023	0.004	0.005	0.049	**0.003**	0.032
Houston	Inference (s)	50.89	28.36	**14.02**	117.48	54.17	18.42	206.54	70.35	60.52	47.05
	Params (M)	–	**0.05**	0.403	50.652	0.651	0.148	0.136	1.044	0.131	1.119
	FLOPs (G)	–	0.28	0.006	0.362	0.032	0.004	0.005	0.026	**0.003**	0.04
Trento	Inference (s)	2.75	2.01	**1.84**	16.61	7.22	2.59	28.64	7.91	9.84	6.43
	Params (M)	–	**0.078**	0.402	53.382	0.306	0.153	0.136	0.754	0.131	0.961
	FLOPs (G)	–	0.022	0.006	0.306	0.014	0.006	0.005	0.053	**0.003**	0.027
LongKou	Inference (s)	**3.91**	15.18	4.97	42.51	66.82	5.9	70.21	27.13	31.04	22.22
	Params (M)	–	0.298	0.404	64.797	1.708	0.153	**0.136**	1.483	0.304	1.57
	FLOPs (G)	–	0.093	0.006	0.538	0.06	**0.004**	0.005	0.114	0.008	0.076

improvements with reasonable FLOPs and inference times, thereby offering a practical solution for high-precision HSI classification.

VI. CONCLUSION

This article introduces FS-CrossMamba, a unified framework aimed at addressing the inherent difficulties of subtle interclass differences and substantial spectral redundancy in HSIC. Unlike conventional Mamba architectures, which compute state transition parameters solely from the input sequence, the proposed method employs a novel dual-branch cross-domain interaction scheme to embed frequency-domain priors (amplitude and phase) and attention-refined spectral information directly into the SSM dynamic modeling process. This integrated design enables the network to disentangle discriminative material signatures from redundant background content while preserving topological consistency in the presence of spatial fragmentation. Comprehensive experiments on five benchmark datasets show that FS-CrossMamba reliably surpasses leading CNN-, Transformer-, and Mamba-based approaches, producing visually coherent classification maps with improved structural integrity.

Currently, frequency priors are applied only within the spatial branch, while the spectral branch relies on attention alongside the standard scanning scheme. We adopted this design to retain Mamba's linear-time computational complexity and to avoid the latency that would arise from processing high-dimensional spectral sequences directly in the frequency domain. As future work, we aim to explore lightweight spectral–frequency approaches, such as 1-D wavelet transforms, to explicitly attenuate noise without sacrificing fast inference. In addition, we plan to investigate cross-dataset generalization settings, such as training on IP and evaluating on Houston2013 or WHU-HI-LongKou, to further assess and enhance the robustness of FS-CrossMamba under significant domain shifts across different sensors and scene distributions.

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